



WEEK 19 – Changing Linear Graphs

Grade: Intermediate (9) **Unit:** Linear relations

Curriculum Expectations

MPM 1D/MFM 1P: describe the effects on a linear graph and make the corresponding changes to the linear equation when the conditions of the situation they represent are varied

SEL: make connections between math and everyday contexts to help them make informed judgements and decisions

Activity

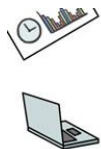
- 1) For this activity, you will be given a series of graphs that represents specific linear relations, and you will need to redraw the line based on a specific criteria that will then be changed
- 2) First, you will analyze the graph as it is displayed with the original description and the changed descriptions
- 3) You will then determine, based on the changed definition, if you need to make a change to the y-intercept of the line, or if you need to change the slope of the line itself. Jot this beside the definitions
- 4) Having figured out the required changes, you will now resketch the line with the required changes to make it fit the new criteria.

Check for Understanding

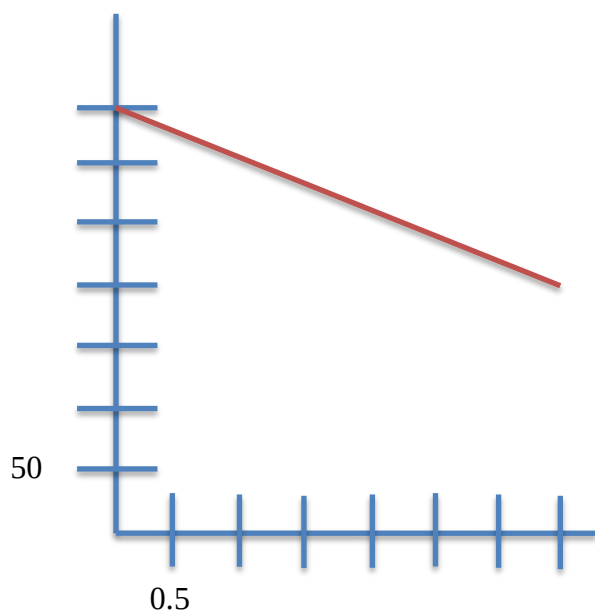
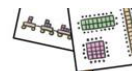
I understand how linear relations represent real life relationships
I can make the required changes to linear relations based on changes to their formulas
I understand the relevance of graphs in my everyday life

Materials

Recording sheet (attached below), pencil, information sheet below or internet access, calculator

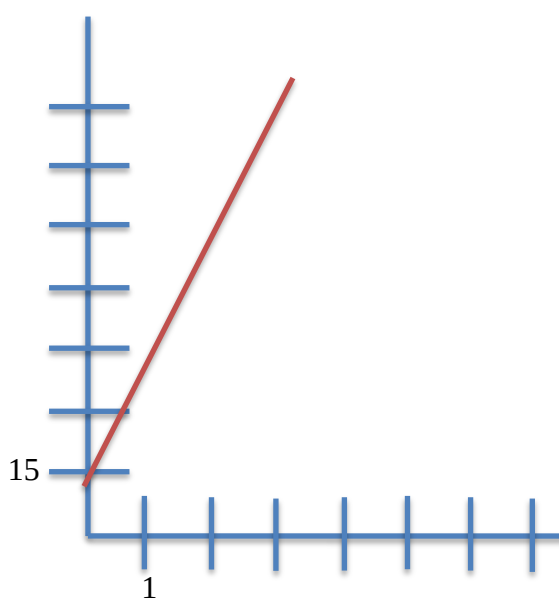


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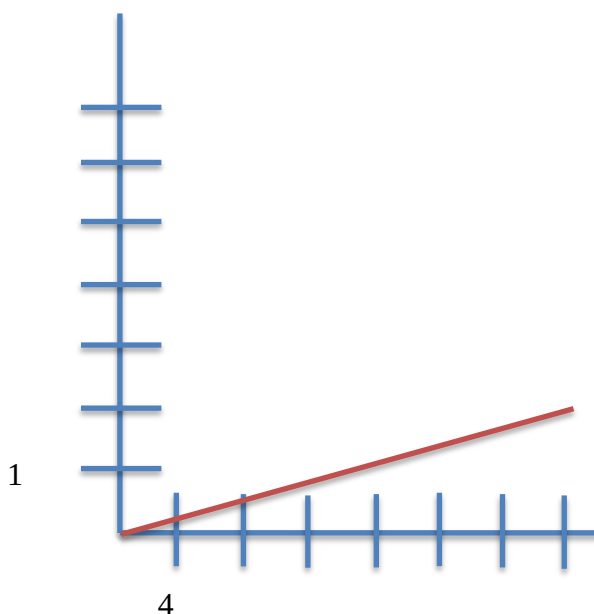
The amount of money in a chequing account after hours in a mall

Change: There is only \$200 in the chequing account at start



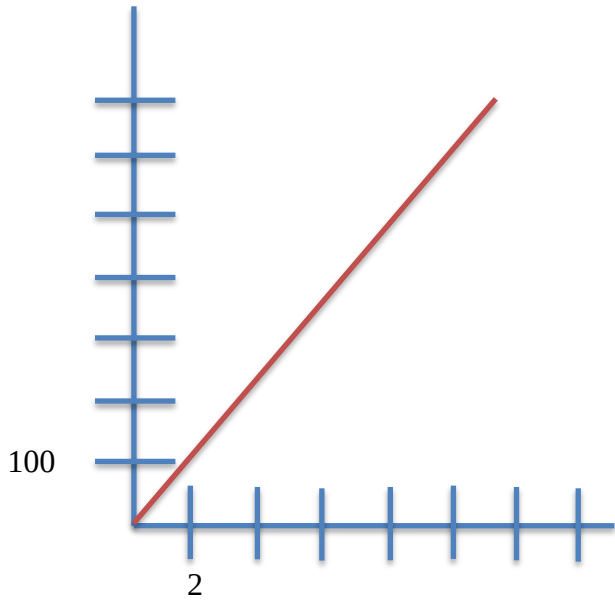
Heat of water in $^{\circ}\text{C}$ in a pot on the stove in minutes

Change: The water increases at 45° per minute



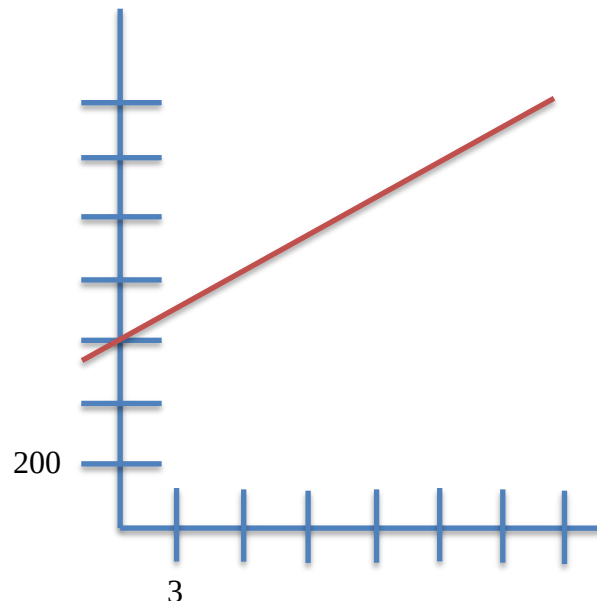
The rate of interest on a Guaranteed Investment Certificate (GIC) per 4 months

Change: The bank promises an initial lump sum interest of 1% with a 2% yearly interest rate



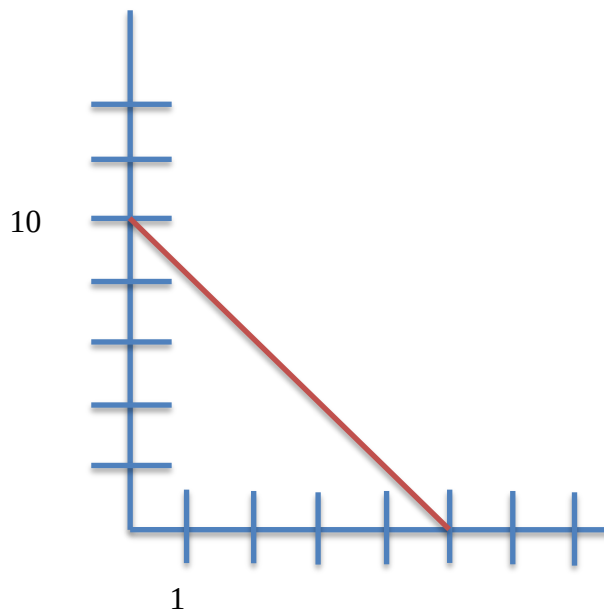
The rate of the cost of a cell phone plan per month with a free phone

Change: The cost of a cell phone plan with a discounted phone at \$100 and \$75



The cost of a cell phone plan per month on a bring your own phone plan

Change: The phone costs \$500 up front at the cost of \$50 per month



The height of a thrown ball in feet from the ground per second

Change: The ball falls at 5ft per second